IOQM · GRADE 9 MATHEMATICS

Graphs, Colouring & Incidence Counting

Diagnostic and benchmark teacher companion

OBJECTS → RELATIONS → GRAPH → INVARIANT → CHECK

Teacher-only companion · independent answer and misconception diagnostics

Teacher Diagnostic Key — Graphs, Colouring and Incidence Counting

Recognition and First-Line Lab

1. B. Vertices are people; edges are the communicating pairs.
2. B. `sum degrees=2|E|`.
3. B. These are proper vertex colourings because adjacency forbids equal colours.
4. A. The endpoints of the missing edge are non-adjacent and may share a colour.
5. B. Cyclic wrap-around can create extra conflicts between the final and initial positions.
6. A. Halving is valid only if every unordered pair was counted exactly twice.
7. A. If only crossings/concurrencies matter, incidence structure is the cheaper representation.
8. B. Turn, history and remaining options can matter in a game state.
9. Example: "Each student is a vertex; two vertices are joined exactly when those students are friends."
10. $2|E| = 4+3+3+2+2+2 = 16$, so $|E| = 8$.
11. $5 \cdot 4 \cdot 3$.
12. Join two cycle vertices exactly when their cyclic distance is 1 or 2.
13. Vertex = one board square; edge = one unordered pair of squares connected by a legal knight move.
14. Count `{(line,point): the selected point lies on the line}`.
15. Fix one vertex and inspect the colours of its four incident edges.
16. At minimum record the player to move and the current remaining/legal configuration; resource counts/history may also matter if they change future moves.

Practice and Transfer Bank

1. Degree sum `10`, so 5 edges.
2. No. The degree sum would be 11, but every graph has even degree sum.
3. First vertex 3 choices, then 2 choices at each of the next three vertices: $3 \cdot 2^3 = 24$.
4. $4 \cdot 3 \cdot 2 = 24$.
5. Let the missing edge be `CD`. Colour `A`: 3; `B`: 2; `C`: 1; `D`: 1 because it may equal C but must differ from A and B. Total 6.
6. $K_{3,3}$ has $3 \cdot 3 = 9$ edges. Degree sum is six vertices of degree 3, total 18, hence 9 edges.
7. Knight edges on an `m x n` board: `2(m-1)(n-2)+2(m-2)(n-1)`. For `4 x 4`: $12 + 12 = 24$.
8. $3 \cdot 4 = 12$ edges; degree sum $3 \cdot 4 + 4 \cdot 3 = 24$, hence 12.
9. For a 5-cycle and 3 colours: $(3-1)^5 - (3-1) = 32 - 2 = 30$.
10. Degree sum 12, so 6 edges.
11. Count line-point incidences: $8 \cdot 3 = 24$. Each selected point is counted twice, so there are 12 points.
12. A `3 x 2` knight rectangle has two knight edges. There are 12 placements of each orientation on a `5 x 5` board, so $24 + 24 = 48$.
13. The number of odd-degree vertices must be even because the total degree sum is even. Hence exactly one odd-degree vertex is impossible.
14. Once the first two colours are distinct, the third is forced to be the remaining colour, and the pattern repeats with period 3. Since 12 is divisible by 3, any ordered choice of distinct first two colours works: $3 \cdot 2 = 6$.
15. Fix a vertex of K6. Among its 5 incident edges, at least 3 have one colour, say red, to vertices A,B,C. If any of AB,BC,CA is red, there is a red triangle with the fixed vertex; if none is red, all three are blue, giving a blue triangle ABC.
16. Total membership incidences $10 \cdot 6 = 60$. Each student contributes 3 incidences, so there are $60/3 = 20$ students.
17. 48. The graph is K4 minus one edge: $4 \cdot 3 \cdot 2 \cdot 2$.

18. 48, by the 3 x 2 / 2 x 3 rectangle count or directed-degree count divided by 2.
19. 12. Every vertex must have exactly two red and two blue incident edges, so the red graph is a labelled 5-cycle; there are $(5-1)!/2 = 12$ such cycles.
20. 19. Equal-coloured vertices must be separated by at least five cyclic steps. For $n = 19$, six colour classes cover at most $6 \cdot floor(19/5) = 18$ vertices, impossible. Explicit valid cyclic patterns exist for 20 through 24 and appending a full five-colour block extends them to every larger n; hence 19 is largest non-colourful.
21. 392. Only the two alternating vertex triples can have all three sides among the nine diagonals. Inclusion-exclusion on those two disjoint all-blue triangles gives $2^9 - 2 \cdot 2^6 + 2^3 = 392$.
22. 9 regions. There are four segments and four cross-family intersections, with no triple concurrency. Region count is $1 + 4 + 4 = 9$.

Mixed Mastery Test

1. Degree sum 14, so 7 edges.
2. No. The degree sum is 19, which is odd.
3. $4 \cdot 3^4 = 324$.
4. Proper colourings of C4 with 3 colours: $(3-1)^4 + (3-1) = 16 + 2 = 18$.
5. $4 \cdot 3 \cdot 2 \cdot 2 = 48$.
6. 24 unordered knight-move pairs.
7. $10 \cdot 6/3 = 20$ students.
8. No. Three named colours and “every three consecutive positions distinct” force a period-3 pattern. A cycle closes only when its length is divisible by 3; 8 is not.
9. Same K6 proof as Practice 15: among five incident edges, three share a colour; their other endpoints force either a triangle in that colour or a triangle in the other colour.
10. Line-point incidences: $12 \cdot 4 = 48$; each point contributes 3 incidences, so 16 points.
11. Outdegree only lists immediate options. Winning strategy can depend on whose turn it is, which position/resources remain, terminal conditions and any history-dependent restrictions.
12. Proper colourings of C7 with 3 colours: $(3-1)^7 - (3-1) = 128 - 2 = 126$.

Historical anchor audit

`IOQM-2025-Q08 = 48.` - exact model is K4 with edge BD removed; - sequential legal choices are `4,3,2,2`.

`IOQM-2025-Q29 = 19.` - conflict graph is the fourth power of the cycle; - equal colours require cyclic separation at least 5; - six colours cannot cover 19 vertices because each class has size at most 3; - explicit base colourings establish all $n \geq 20$.

`IOQM-2024-Q09 = 48.` - each knight edge is one diagonal pair of a 3 x 2 or 2 x 3 rectangle; - 12 placements of each orientation, 2 edges each.

`IOQM-2024-Q19 = 12.` - triangle avoidance forces degree 2 in each colour at every vertex; - the red subgraph is a labelled 5-cycle; its complement is the blue 5-cycle.

`IOQM-2023-Q16 = 94.` - nine diagonals have 392 valid red/blue colourings after excluding the two possible all-blue alternating triangles; - requested digit-square sum is $3^2 + 9^2 + 2^2 = 94$.

`IOQM-2023-Q22 = 77.` - region count 9 requires intersection contribution 4; - `(2,2,0)` family distribution contributes 300 selections; - `(2,1,1)` requires exactly one triple concurrency; discrete Ceva gives 13 concurrent triples, contributing $13 \cdot 3 \cdot 4 = 156$; - total 456, digit-square sum $16 + 25 + 36 = 77$.

Diagnostic map

- graph terminology used before defining the model -> ask what one vertex and one edge mean;
- unrestricted colour count used -> list adjacency restrictions before multiplying;
- divide-by-two done automatically -> demand proof that every desired object was counted exactly twice;
- cyclic problem treated as linear -> inspect wrap-around conflicts;
- degree sum odd but accepted -> use handshaking parity;
- incidence double count has two different object sets -> explicitly define the incidence pair before equating counts;
- Ramsey problem attacked by raw enumeration -> fix one vertex and force local colour degrees;

- geometric surface attacked with coordinates although only crossings matter -> move to incidence structure;
- static graph count used for an adversarial game -> add player-to-move and future-state information.

Teacher Key — COMB-02 Benchmark Assimilation Lab

This key supports $07_{B}enchmark_{A}ssimilation_{L}ab.md$. It is separate from the main Teacher Diagnostic Key so the benchmark-specific diagnostic can be checked independently.

A. RECONNECT

1. Define the model: what one vertex represents and exactly when two vertices are joined by an edge.
2. Every edge contributes 1 to the degree of each of its two endpoints, so `sum degrees = 2|E|`, which is even.
3. Unrestricted colouring permits any colour at any vertex. Proper colouring requires every adjacent pair to receive different colours.
4. A cycle has wrap-around adjacency: the final positions can be adjacent to the initial positions, so a linear count can violate the closure condition.
5. Prove that every desired unordered pair occurs exactly twice in the directed count, once from each endpoint, and that no loops/exceptional objects have a different multiplicity.
6. Test an incidence/intersection graph: retain which cevians cross or concur and discard irrelevant metric data.
7. Fix one vertex and inspect the colours of its incident edges; pigeonhole/local forcing often determines the next structure.
8. At minimum record whose turn it is and the current legal configuration/resources. Add history when it changes future legal moves or terminal conditions.

B. Error laboratory

Error 1

A geometric drawing is not the model. First write, for example, `vertex = one person` and `edge = one communicating pair`; only then choose a drawing convenient for reasoning.

Error 2

3^4 counts unrestricted assignments. If adjacency matters, list the graph restrictions first and count only legal colour choices.

Error 3

A total of 40 directed moves implies 20 edges only if each desired unordered edge is counted exactly twice. Verify that condition before halving.

Error 4

For a cycle, include the edge/conflict between the last and first positions from the start. A path count cannot simply ignore closure.

Error 5

Double counting requires one identical set. Define, for example, `I={(line,point): selected point lies on selected line}` and count `I` by lines and then by points.

Error 6

For a two-colouring of K5 or K6, raw enumeration hides the invariant. Fix one vertex; pigeonhole forces several incident edges of one colour, and the edges among their other endpoints force a monochromatic triangle or the complementary structure.

C. ADOPT

1. First lines: `vertex = one student`; `edge = one unordered communicating pair`.
2. `sum degrees = 4+4+3+3+2+2 = 18`; therefore $2|E| = 18$, so $|E| = 9$.
3. A path of length 5 has 6 vertices. First vertex: 4 choices; each subsequent vertex: 3 choices, so the count begins $4 \cdot 3^5$.
4. Model `C7`. A standard cycle-colouring route with $q = 3$ gives $(q-1)^7 - (q-1) = 2^7 - 2 = 126$; alternatively carry a path colouring with an explicit closure check.
5. Count knight edges by rectangle placements or degrees. For an `m x n` board, `2(m-1)(n-2)+2(m-2)(n-1)`. With $m = 6, n = 5$: $2 \cdot 5 \cdot 3 + 2 \cdot 4 \cdot 4 = 62$ unordered knight edges.
6. Count incidences. By lines: $10 \cdot 4 = 40$. Each selected point lies on two lines, so $2P = 40$, giving $P = 20$.
7. Fix a vertex of K6. Of its five incident edges, at least three have one colour, say red, to `A,B,C`. If any of `AB,BC,CA` is red, it forms a red triangle with the fixed vertex; if none is red, all three are blue and `ABC` is a blue triangle.
8. Build the cycle conflict graph in which two polygon vertices are adjacent exactly when their cyclic distance is at most 3. The colouring rule becomes ordinary proper colouring of this graph.

D. TRANSFER

1. Translate "number of games played by each team" into vertex degrees. Before constructing a schedule, check handshaking parity and degree bounds such as $0 \le d(v) \le n-1$ in a simple schedule graph.
2. Create a graph on the cyclic seats and connect two seats when cyclic distance is at most 2. Then badge assignments are proper colourings of that conflict graph.
3. Route A: count legal jumps by displacement/rectangle classes. Route B: sum local degrees and divide by 2. They must agree only when both count the same unordered legal-jump edge set and the degree sum counts each edge exactly twice.
4. Define `I={(S,x): element x belongs to selected set S}`. Count $|I|$ by summing set sizes and again by summing, over elements, the number of sets containing each element.
5. Preserve which cevians meet, the multiplicity of each concurrence, endpoint/family information needed to determine allowable crossings, and boundary incidence. Discard exact lengths/angles when they cannot change that incidence pattern.
6. Example: "How many legal moves are available from each square of a fixed board?" can be a static move-graph degree-count problem even though the word *move* appears. It becomes a game-state problem only when players make adversarial sequential choices and future legality/winning depends on the evolving state.

E. Six-question assimilation test — diagnostic rubric

A strong response should contain all six components:

1. a specific relation clue rather than the chapter name;
2. an explanation of what the proposed graph/incidence representation preserves;
3. a reusable future recognition cue;
4. a genuine near-neighbour requiring a different route;
5. two mathematically productive opening lines;
6. a changed-surface example with the same invariant and a valid route.

Diagnostic tags:

- `recognition`: relation pattern not identified;
- `modelling`: vertex/edge/incidence meanings are ambiguous or wrong;
- `representation`: a more complex or lossy representation chosen;
- `counting`: multiplication/addition/complement applied to the wrong object set;
- $double_count$: halving/equating counts without multiplicity proof;

- $cyclic_closure$: wrap-around restriction omitted;
- `forcing`: Ramsey/local inevitability replaced by unstructured enumeration;
- `boundary`: static graph confused with adversarial game state;
- `transfer`: surface and mathematical invariant both changed.

Independent QA disposition

All deterministic counts and route claims above were recomputed after the learner lab was authored. No answer was copied from the learner-facing text.

`BENCHMARK_LAB_KEY_STATIC_CHECK = PASS`.